DATA 266 — Homework 10

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SID4 = 0670 | SEED = 670 | SLICE = 670 | HP_ID = 4 | CLS_A = 0 | CLS_B = 5

Notebook: `rag_rerank_memory.ipynb`. Dataset: `ccdv/arxiv-summarization`, `document`, first 100 train articles.

Same SID/SEED as the earlier homework. HP_ID is reported only.

1. Data and preprocessing

I loaded the Hugging Face dataset `ccdv/arxiv-summarization` with the `document` config and kept `train[:100]`. Each row has a long `article` and a short `abstract`. Article length was about 3k to 161k characters (mean about 32k). I cleaned `@xmath` / `@xcite` tokens and left the text otherwise intact.

2. Two chunking strategies

Strategy	Splitter	Size / overlap	Chunks
Fixed-size	`CharacterTextSplitter`	500 / 50	7,220
Recursive	`RecursiveCharacterTextSplitter`	500 / 50	7,224

That matches Level 1 and Level 2 from the lecture. Both indexes use the same embedding model so the comparison is on the chunk boundaries.

3. Embeddings and ChromaDB

I embedded with `sentence-transformers/all-MiniLM-L6-v2` and stored one Chroma collection per strategy under `chroma_store/`. Indexing both collections took 3.02 minutes on my Mac.

4. Baseline retrieval and reranking

Queries:

- **Q1:** learning rates for regularized kernel based methods for additive models (gold doc 0)
- **Q2:** positive linear magnetoresistance in topological insulator surface states (gold doc 20)

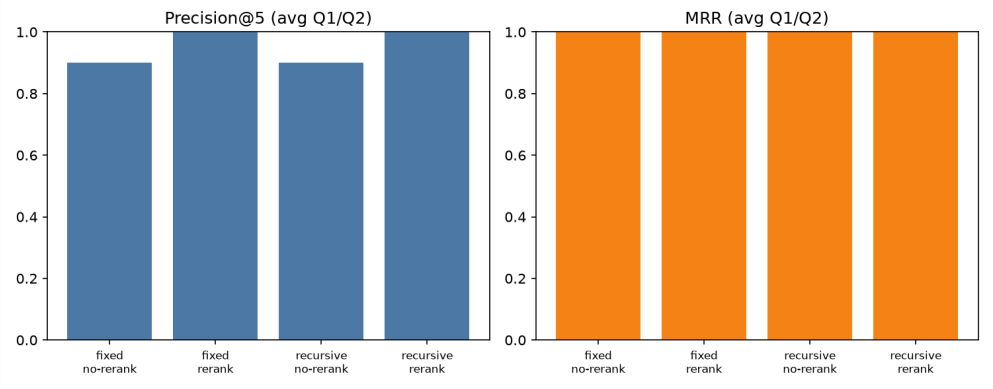
Baseline retrieval returns embedding top-5. Reranked retrieval returns embedding top-20, then `cross-encoder/ms-marco-MiniLM-L-6-v2` keeps top-5.

A chunk counts as relevant if its source `doc_id` is in the gold set for that query.

Setting	Precision@5	Recall@5	MRR
fixed / Q1 / no rerank	1.000	1.000	1.000
fixed / Q1 / rerank	1.000	1.000	1.000
fixed / Q2 / no rerank	0.800	1.000	1.000
fixed / Q2 / rerank	1.000	1.000	1.000
recursive / Q1 / no rerank	1.000	1.000	1.000
recursive / Q1 / rerank	1.000	1.000	1.000
recursive / Q2 / no rerank	0.800	1.000	1.000
recursive / Q2 / rerank	1.000	1.000	1.000

Average over Q1 and Q2:

Setting	Precision@5	MRR
fixed, no rerank	0.900	1.000
fixed, rerank	1.000	1.000
recursive, no rerank	0.900	1.000
recursive, rerank	1.000	1.000



Chunk and rerank comparison

Reranking raised average Precision@5 from 0.900 to 1.000. On Q2 the no-rerank list kept document 71 once; after reranking all five hits were document 20. Fixed and recursive were almost the same on these two queries, so I am not claiming a chunking winner from this small set.

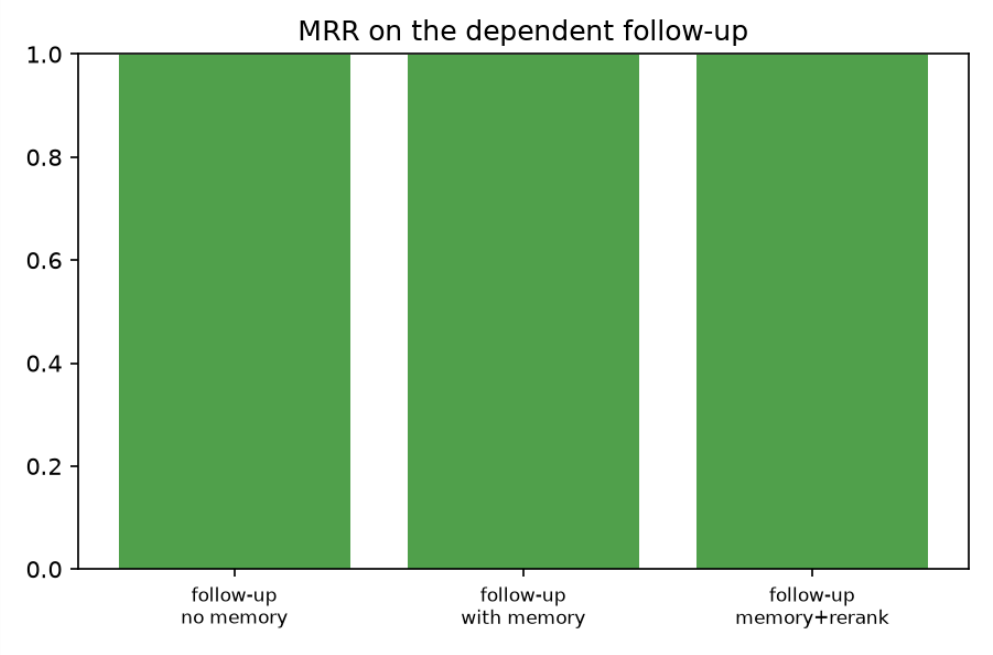
5. Conversational memory

I stored the first user question and a short answer sketch. The follow-up was:

> What learning rates does the paper report?

Without memory that bare string goes to the retriever. With memory I rewrite it using the previous user turn so it stays about additive-model kernel methods.

Setting	Precision@5	Recall@5	MRR	Top doc ids
follow-up, no memory	0.800	1.000	1.000	0, 0, 0, 0, 51
follow-up, with memory	1.000	1.000	1.000	0, 0, 0, 0, 0
follow-up, memory + rerank	1.000	1.000	1.000	0, 0, 0, 0, 0



Memory comparison

MRR was already 1.000 without memory because document 0 was first. Memory still helped: Precision@5 went from 0.800 to 1.000 by removing the off-topic document 51.

Conclusion

Chroma + MiniLM was enough to retrieve the right paper on both content queries. The cross-encoder cleaned the Q2 top-5. Recursive and fixed chunking looked similar here. Conversational memory mattered on the dependent follow-up by keeping every top-5 chunk on the same paper.